

Combined Oxidative Phosphorylation Defect Type 26 (COXPD26): A Comprehensive Disease Characterization

Disease: Combined Oxidative Phosphorylation Defect Type 26 (COXPD26) — the TRMT5-related mitochondrial disease **Key identifiers:** OMIM 616539 · MONDO:0014684 · Gene: *TRMT5* (HGNC:23141) **Category:** Mendelian, autosomal recessive **Note:** This entry concerns the TRMT5-related disorder and is distinct from COXPD30 / *TRMT10C* (OMIM 616974).

Summary

Combined Oxidative Phosphorylation Defect Type 26 (COXPD26; OMIM 616539; MONDO:0014684) is an **ultra-rare autosomal recessive primary mitochondrial disease** caused by biallelic loss-of-function variants in *TRMT5*, the nuclear-encoded tRNA methyltransferase that installs the **N1-methylguanosine (m1G37)** modification on mitochondrial tRNAs 3' to the anticodon. Loss of TRMT5 activity produces **hypomodification of G37 in mitochondrial tRNAs** — most prominent in skeletal muscle — which impairs mitochondrial translation and results in a **combined deficiency of multiple respiratory-chain complexes** accompanied by lactic acidosis. The disease was first defined molecularly by Powell et al. in 2015 through whole-exome sequencing of two unrelated patients ([PMID: 26189817](#)).

The clinical picture is **highly variable and multisystem**. The originally reported extremes ranged from a child with failure to thrive and hypertrophic cardiomyopathy to an adult with a lifelong history of exercise intolerance. A subsequently reported Chinese case broadened the spectrum to include developmental delay, gastrointestinal dysmotility, hypotonia/muscle weakness, neuropathy, spastic diplegia, seizures, and novel renal and hepatic involvement ([PMID: 35109800](#)). The disorder preferentially injures high-energy, post-mitotic tissues (skeletal muscle, heart, central and peripheral nervous system), consistent with a defect in mitochondrial energy production.

There is **no disease-specific or curative therapy**. Management follows general primary mitochondrial disease principles — supportive, organ-directed care, and metabolic cofactor supplementation (e.g., CoQ10, riboflavin), with generally modest efficacy. Because etiology is purely genetic, prevention rests on **genetic counseling, carrier detection, and reproductive options**. Fewer than roughly ten patients have been reported worldwide; a gnomAD-based estimate places the null-allele carrier frequency near **1 in 350**, though true disease incidence is higher because hypomorphic missense alleles also cause disease.

Section-by-Section Characterization

1. Disease Information

COXPD26 is a nuclear-gene mitochondrial translation disorder. It belongs to the "combined oxidative phosphorylation deficiency" (COXPD) family of OMIM phenotypes — disorders in which **multiple** respiratory-chain complexes are simultaneously deficient, typically due to defects in the mitochondrial protein-synthesis machinery rather than in a single structural OXPHOS subunit.

- **What it is:** An autosomal recessive multisystem mitochondrial disease caused by biallelic *TRMT5* variants, characterized biochemically by lactic acidosis and combined respiratory-chain-complex deficiency, and molecularly by mt-tRNA m1G37 hypomodification.
- **Key identifiers:** OMIM **616539**; MONDO **:0014684**; Gene *TRMT5* (HGNC:23141, NCBI Gene 57570, Ensembl ENSG00000126814, UniProt Q32P41). A specific Orphanet code is not firmly established (ultra-rare); ICD-11 maps broadly to mitochondrial disease (e.g., 5C53.0 / relevant metabolic mitochondrial category); MeSH has no dedicated descriptor (indexed under "Mitochondrial Diseases" / "OXPHOS deficiency").
- **Synonyms / alternative names:** "Combined oxidative phosphorylation deficiency 26"; "COXPD26"; "TRMT5-related mitochondrial disease"; "TRMT5 deficiency"; "mitochondrial tRNA m1G37 methyltransferase deficiency."
- **Information source type:** Disease-level knowledge is **aggregated from individual published case reports** (a handful of patients) plus enzymology/mechanistic studies — not from large EHR cohorts or registries. Findings are thus case-based.

2. Etiology

- **Disease causal factors:** Purely **genetic** — biallelic (compound heterozygous or homozygous) loss-of-function variants in *TRMT5* (PMID: 26189817). No infectious, environmental, or primary toxic cause.
- **Genetic risk factors:** The causal variants themselves; disease requires two defective alleles. No susceptibility loci or validated modifier genes reported.
- **Environmental risk factors:** None established as causal. By analogy to primary mitochondrial disease, **catabolic stressors** (infection, fever, fasting, dehydration, general anesthesia, certain mitotoxic drugs) may act as **symptom precipitants** — an inferred, not demonstrated, COXPD26-specific interaction.
- **Protective factors:** None known (no protective alleles, no disease-modifying diet/exposure).
- **Gene–environment interactions:** Inferred only — metabolic decompensation triggered by intercurrent illness, as seen generically in mitochondrial disease.

3. Phenotypes

The spectrum is broad and multisystem. In the original report *"one presented in childhood with failure to thrive and hypertrophic cardiomyopathy, and the other was an adult with a life-long history of exercise intolerance"* (PMID: 26189817). The Chinese case added *"developmental delay, gastrointestinal dysfunction, shortness of breath, exercise intolerance, hypotonia and muscle weakness, neuropathy, and spastic diplegia"* plus renal and hepatic involvement and seizures (PMID: 35109800).

Phenotype	HPO term (suggested)	Type	Onset	Notes
Lactic acidosis	HP:0003128	Laboratory abnormality	Variable	Biochemical hallmark
Combined OXPHOS/ respiratory-chain deficiency	HP:0011922	Laboratory abnormality	Variable	Multiple complexes in muscle
Hypertrophic cardiomyopathy	HP:0001639	Clinical sign	Childhood	Severe end of spectrum
Failure to thrive	HP:0001508	Clinical sign	Childhood	
Exercise intolerance	HP:0003546	Symptom	Childhood–adult	Can be isolated adult presentation
Muscle weakness	HP:0001324	Clinical sign	Variable	Myopathy
Hypotonia	HP:0001252	Clinical sign	Variable	
Global developmental delay	HP:0001263	Clinical sign	Childhood	
Seizures	HP:0001250	Clinical sign	Childhood	
Peripheral neuropathy	HP:0009830	Clinical sign	Variable	
Spastic diplegia	HP:0001264	Clinical sign	Childhood	
Gastrointestinal dysmotility	HP:0002579	Clinical sign	Variable	
Abnormal liver function	HP:0001392	Clinical sign	Variable	Novel (Chinese case)
Renal involvement	HP:0000077	Clinical sign	Variable	Novel (Chinese case)

- **Severity/progression:** Variable; ranges from mild (isolated adult exercise intolerance) to severe childhood cardiomyopathy/failure to thrive. Progressive in severe cases.
- **Frequency among affected individuals:** Qualitative only — too few patients for reliable percentages.

- **Quality-of-life impact:** Not formally quantified (no EQ-5D/SF-36 data). Severe childhood disease with cardiomyopathy and developmental delay is highly disabling; isolated adult exercise intolerance is comparatively mild.

4. Genetic / Molecular Information

- **Causal gene:** *TRMT5* — HGNC:23141, NCBI Gene 57570, Ensembl ENSG00000126814, **chromosome 14q23.1** (GRCh37 chr14:60,971,441–60,981,694, minus strand), UniProt Q32P41. Encodes a **class I-like (Rossmann-fold) SAM-dependent methyltransferase**; "*TRMT5* encodes a mitochondrial protein with strong homology to members of the class I-like methyltransferase superfamily" (PMID: 26189817).
- **Pathogenic variants (transcript NM_020810.3):** Pathogenic missense **c.1156A>G (p.Met386Val)**; likely-pathogenic truncating **c.259A>T (p.Arg87Ter)** and **c.267_270delinsCTG (p.Ala89_Phe90insTer)**; reported case variants **c.665T>C (p.Ile222Thr)** and the compound-heterozygous set **c.881A>C (p.Glu294Ala)**, **c.1218G>C (p.Gln406His)**, **c.1481C>T (p.Thr494Met)** (PMID: 35109800); plus LoF alleles **c.205_208del (p.Glu69fs)**. Variant classes span missense (predominant, hypomorphic), nonsense, frameshift, in-frame indel-with-stop, and splice/UTR.
- **Classification:** Per ACMG/AMP; ClinVar (~299 records) is dominated by VUS and likely-benign calls, so functional evidence is often needed.
- **Allele frequency:** Individual pathogenic alleles are ultra-rare in gnomAD; summed predicted-LoF allele frequency ≈ 0.00143 .
- **Somatic vs germline:** All germline.
- **Functional consequence:** **Loss of function** in every case — convergent loss of m1G37 methyltransferase activity; no gain-of-function or dominant-negative described. LoF proven by rescue with wild-type cDNA (PMID: 26189817).
- **Constraint:** gnomAD pLI = 0.05, LOEUF = 0.588, missense Z = 0.10 — tolerant of heterozygous LoF, as expected for a recessive gene.
- **Modifier genes / epigenetics / chromosomal abnormalities:** None reported.

5. Environmental Information

No environmental, lifestyle, or infectious factor causes COXPD26. It is a monogenic disorder. Catabolic/metabolic stress may precipitate decompensation (inferred). No toxins, occupational exposures, dietary factors, or pathogens are implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic loss-of-function TRMT5 variants (missense hypomorph and/or truncating)
 - | leads to
2. Reduced/absent TRMT5 (m1G37) methyltransferase activity in the mitochondrial matrix
 - | results in
3. Hypomodification of guanosine-37 (loss of m1G37) on mitochondrial tRNAs
 - | (most prominent in skeletal muscle)
 - | leads to
4. Impaired mitochondrial translation of mtDNA-encoded OXPHOS subunits
 - | (loss of decoding fidelity / frameshift-prevention function of m1G37)
 - | results in
5. Combined deficiency of multiple respiratory-chain complexes
 - | branches to ↓ (downstream, partly inferred from related tRNA-37 defects)
 - ↳ reduced mitochondrial membrane potential + reduced ATP production → energy failure
 - ↳ accumulation of reducing equivalents → elevated lactate → LACTIC ACIDOSIS
 - ↳ increased reactive oxygen species (ROS) → oxidative stress → autophagy / cell injury
 - | leads to (preferential injury of high-energy, post-mitotic tissues)
6. Organ dysfunction:
 - ↳ skeletal muscle → myopathy, exercise intolerance, hypotonia
 - ↳ heart → hypertrophic cardiomyopathy
 - ↳ CNS → developmental delay, seizures, spastic diplegia
 - ↳ peripheral nerve → neuropathy
 - ↳ GI / liver / kidney → dysmotility, hepatic & renal involvement
 - | culminates in
7. Variable multisystem clinical disease (childhood cardiomyopathy/FTT ↔ adult exercise intolerance)

Molecular detail. TRMT5 catalyzes **SAM-dependent methyl transfer to N1 of guanosine 37** on mt-tRNAs: "*Trm5 is a eukaryal and archaeal tRNA methyltransferase that catalyzes methyl transfer from S-adenosylmethionine (AdoMet) to the N(1) position of G37 directly 3' to the anticodon*" (PMID: 20980671). m1G37 preserves reading frame and decoding fidelity: "*Enzymes of the Trm5 family catalyze methyl transfer from S-adenosyl methionine (AdoMet) to the N¹ of G37 to synthesize m¹G37-tRNA as a critical determinant to prevent ribosome frameshift errors*" (PMID: 23887145); structural work confirms m1G37 stabilizes codon–anticodon pairing (PMID: 33016876). In patients, "*Mutations in TRMT5 were associated with the hypomodification of a guanosine residue at position 37 (G37) of mitochondrial tRNA; this hypomodification was particularly prominent in skeletal muscle*" (PMID: 26189817).

Downstream cascade (inferred from a closely related position-37 mt-tRNA defect): *"The aberrant tRNA metabolism resulted in the impairment of mitochondrial translation, respiratory deficiency, decreasing membrane potentials and ATP production, increasing production of reactive oxygen species and promoting autophagy"* (PMID: 33398350).

Upstream vs downstream summary:

Layer	Event	Status
Upstream (cause)	Biallelic <i>TRMT5</i> LoF variants	Demonstrated
Molecular lesion	Loss of m1G37 on mt-tRNAs	Demonstrated (patients + RNAi)
Proximal effect	Impaired mitochondrial translation	Demonstrated
Biochemical readout	Combined RC-complex deficiency; lactic acidosis	Demonstrated
Downstream	↓ ATP, ↓ $\Delta\psi_m$, ↑ ROS, autophagy	Inferred (analogous tRNA-37 defects)
Clinical	Multisystem organ dysfunction	Demonstrated (case reports)

- **GO terms:** mitochondrial translation (GO:0032543); tRNA methylation (GO:0030488); mitochondrion (GO:0005739); mitochondrial matrix (GO:0005759).
- **CL terms:** skeletal muscle cell (CL:0000188); cardiac muscle cell / cardiomyocyte (CL:0000746); neuron (CL:0000540).
- **CHEBI terms:** S-adenosyl-L-methionine (CHEBI:15414); N1-methylguanosine (CHEBI:has m1G); lactate (CHEBI:24996).

7. Anatomical Structures Affected

- **Primary organs:** Skeletal muscle (UBERON:0001134), heart (UBERON:0000948 / cardiac muscle UBERON:0002349), brain/CNS (UBERON:0000955).
- **Secondary/additional:** Peripheral nerve (UBERON:0001021), gastrointestinal tract (UBERON:0001555), liver (UBERON:0002107), kidney (UBERON:0002113).
- **Body systems:** Musculoskeletal, cardiovascular, central & peripheral nervous, digestive, hepatic, renal.
- **Tissue/cell level:** Post-mitotic, high-oxidative-demand cells — myocytes, cardiomyocytes, neurons.

- **Subcellular:** Mitochondrial matrix translation apparatus (GO:0005759; GO:0005739). The molecular defect is fundamentally subcellular; organ damage is downstream of energy failure.
- **Localization/lateralization:** Systemic and bilateral/symmetric (metabolic disease), not focal.

8. Temporal Development

- **Onset:** Ranges from **childhood** (failure to thrive, hypertrophic cardiomyopathy) to **adulthood** (isolated exercise intolerance) (PMID: 26189817); the Chinese case had childhood/developmental onset (PMID: 35109800). Onset pattern is typically **chronic/insidious**, with possible acute metabolic decompensation during intercurrent illness.
- **Progression:** Variable — progressive in severe childhood cases; relatively stable in mild adult cases. No formal staging exists.
- **Duration:** Chronic, lifelong.
- **Remission:** No spontaneous remission described. (Note: some *other* mt-tRNA-modification disorders, e.g., TRMU-related, can be reversible — PMID: 21931168 — but reversibility has **not** been reported for COXPD26.)
- **Critical periods:** Early childhood is the highest-risk window for severe presentation; metabolic stress is a period of vulnerability.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive; reported patients are compound heterozygous — *"we identified two unrelated individuals carrying compound heterozygous variants in TRMT5"* (PMID: 26189817); *"This disease is considered to be caused by compound heterozygous mutations in the TRMT5 gene"* (PMID: 35109800).
- **Epidemiology:** Ultra-rare — fewer than ~10 reported patients worldwide. No established Orphanet prevalence.
- **Carrier frequency / incidence (computed):** gnomAD r2.1 contains 44 predicted-LoF *TRMT5* alleles (summed AF ≈ 0.00143), giving a **null-allele carrier frequency** $\sim 2q \approx 0.29\%$ ($\sim 1/350$) and a biallelic-null birth incidence $\sim q^2 \approx 2 \times 10^{-6}$ ($\sim 1/490,000$). This is a **lower bound** — hypomorphic missense alleles also cause disease.
- **Penetrance / expressivity:** Presumed complete penetrance for biallelic LoF; **variable expressivity** (wide phenotype range).

- **Founder effects / consanguinity:** No founder mutation reported; cases span European and East Asian ancestry. Consanguinity would raise homozygous risk (general AR principle).
- **Sex ratio / age distribution:** No sex predilection expected (autosomal); onset childhood–adult.

10. Diagnostics

- **Laboratory:** Elevated blood (and often CSF) **lactate** (lactic acidosis).
- **Muscle biopsy / enzymology:** Combined (multiple-complex) respiratory-chain deficiency; G37 hypomodification most prominent in skeletal muscle — *"this hypomodification was particularly prominent in skeletal muscle"* (PMID: 26189817). **Caveat:** RC enzymology is an insensitive screen for mt-tRNA defects — *"RC enzyme analysis in muscle is not a sensitive test for MM in adults"* (PMID: 19941338).
- **Genetic testing (diagnostic gold standard):** WES/WGS or nuclear mitochondrial-gene panels detecting biallelic *TRMT5* variants; single-gene *TRMT5* sequencing for targeted confirmation. Because most ClinVar entries are VUS, **functional confirmation** (m1G37 methylation assay, yeast complementation) strengthens classification.
- **Imaging/electrophysiology:** Echocardiography (cardiomyopathy), EMG/nerve conduction (neuropathy/myopathy), EEG (seizures), brain MRI as clinically indicated — nonspecific.
- **Differential diagnosis:** Other combined-OXPHOS / mt-tRNA-modification disorders — ACAD9 deficiency (PMID: 33204590), MT-ND5 cardiomyopathy (PMID: 30587702), TRMU-related reversible infantile RC deficiency (PMID: 21931168), MERRF/MELAS. Distinguishing feature: **biallelic nuclear *TRMT5* variants with muscle-predominant mt-tRNA G37 hypomodification**. Explicitly **not** COXPD30/*TRMT10C*.
- **Screening:** No newborn screening exists; cascade carrier testing of relatives once familial variants known.

11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics (too few cases). Severe childhood cardiomyopathy with failure to thrive carries higher mortality risk, as in comparable mitochondrial cardiomyopathies; mild adult disease is compatible with long survival.
- **Morbidity/function:** Highly variable — from mild exercise limitation to severe disability with developmental delay, seizures, and cardiomyopathy.
- **Complications:** Heart failure/arrhythmia (cardiomyopathy), metabolic decompensation during illness, feeding failure, seizures.

- **Prognostic factors:** Age of onset and organ involvement (childhood cardiomyopathy = worse prognosis; isolated adult exercise intolerance = better). Residual TRMT5 activity (allele severity) is a plausible but unproven prognostic determinant.
- **QoL measures:** Not formally assessed.

12. Treatment

- **No disease-specific/curative therapy exists.** Management follows general primary mitochondrial disease practice — *"electron acceptors, enzyme activators, vitamins, coenzymes, free-radical scavengers, dietary measures, and supportive therapy"* (PMID: [11282042](#)), with generally modest efficacy.
- **Cofactor/pharmacotherapy (empiric):** Coenzyme Q10 (CHEBI:46245; NCIT:C1042), riboflavin/vitamin B2 (CHEBI:17015), other B-vitamins, free-radical scavengers, L-carnitine — individualized.
- **Organ-directed/supportive:** Cardiomyopathy management (standard heart-failure care); physical therapy for myopathy/exercise intolerance; nutritional support for GI dysmotility/failure-to-thrive; anticonvulsants for seizures; avoidance of mitotoxic drugs and prolonged fasting; aggressive management of intercurrent illness.
- **Advanced/experimental therapeutics:** No gene therapy, RNA therapy, or targeted therapy in trials for COXPD26. NCIT term for approach: Supportive Care (NCIT:C133435).
- **Pharmacogenomics/personalized medicine:** Genotype-guided cofactor trials plausible but unvalidated; note riboflavin responsiveness in the related ACAD9 deficiency (PMID: [33204590](#)) motivates empiric cofactor trials.

13. Prevention

- **Primary prevention:** Not applicable in the classic sense (monogenic). Prevention centers on **genetic counseling and reproductive planning**.
- **Genetic screening:** Carrier testing of at-risk relatives (cascade testing); prenatal diagnosis or preimplantation genetic testing (PGT) once familial variants are identified.
- **Secondary/tertiary prevention:** Early recognition and organ surveillance (cardiac monitoring), prompt treatment of metabolic decompensation, avoidance of catabolic stress and mitotoxic exposures.
- **Counseling:** Recurrence risk 25% for a couple who are both carriers (AR). No population-based screening program exists (ultra-rare).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** The Trm5/TRMT5 enzyme family is conserved from archaea to humans — *"These results establish conservation in both the catalytic mechanism and overall structure of Trm5 between evolutionarily distant eukaryotic and archaeal species"* (PMID: [23887145](#)). Orthologs: mouse *Trmt5* (NCBI Gene 76357), rat *Trmt5* (362754), zebrafish *trmt5* (564078).
- **Natural disease in other species:** No naturally occurring TRMT5-related disease is documented in companion animals or wildlife (no OMIA entry noted).
- **Comparative biology:** Deep conservation of the m1G37 modification mechanism implies disease mechanisms would be conserved, supporting cross-species modeling.
- **Transmission:** Not applicable (non-infectious, non-zoonotic).

15. Model Organisms

- **Cellular/in vitro:** *TRMT5* RNAi in human cells reproduces G37 hypomodification; re-expression of wild-type cDNA rescues the molecular phenotype (PMID: [26189817](#)).
- **Yeast complementation:** *"The pathogenicity of the detected variants was further confirmed in a heterologous yeast model"* (*S. cerevisiae*) (PMID: [26189817](#)) — a validated system for assessing variant function.
- **Potential whole-animal models:** Conserved orthologs in mouse, rat, and zebrafish exist, but **no published knockout/knock-in animal model** recapitulating the multisystem human disease has been reported.
- **Phenotype recapitulation:** Cellular/yeast models faithfully reproduce the **molecular** phenotype (G37 hypomodification, impaired function). Whole-organism, multisystem recapitulation is an open gap.
- **Applications:** Existing models support variant classification and mechanistic study; iPSC-derived tissues and conditional animal models are the logical next resources.

Mechanistic Model / Interpretation

COXPD26 is a clean example of a **mitochondrial translation disorder**: a single nuclear enzyme deficiency (TRMT5) removes one specific chemical mark (m1G37) from mitochondrial tRNAs, degrading the fidelity and efficiency of mitochondrial protein synthesis, which in turn

cripples **all** thirteen mtDNA-encoded OXPHOS subunits at once — hence a *combined* respiratory-chain deficiency rather than an isolated complex defect. The steps from mutation through m1G37 loss to combined RC deficiency and lactic acidosis are **directly demonstrated in patients and in cell/yeast models**. The specific injurious downstream events (membrane-potential collapse, ATP decline, ROS surge, autophagy) are **inferred** from a mechanistically parallel position-37 mt-tRNA modification disorder ([PMID: 33398350](#)).

The tissue distribution follows an energetics logic: skeletal muscle, heart, and neurons — the most oxidative, least glycolytically flexible, post-mitotic tissues — bear the brunt, explaining myopathy, cardiomyopathy, developmental delay, seizures, and neuropathy. The wide phenotypic range (severe childhood cardiomyopathy vs. isolated adult exercise intolerance) most plausibly reflects **allele severity / residual enzyme activity** and possibly tissue-specific modification thresholds, though this genotype–phenotype relationship remains unproven given the tiny cohort.

Evidence Base

PMID	Title (abbrev.)	Role in this report
26189817	<i>TRMT5 Mutations Cause a Defect in Post-transcriptional Modification of Mitochondrial tRNA...</i>	Landmark / disease-defining. Biallelic <i>TRMT5</i> cause, AR inheritance, combined RC deficiency, m1G37 hypomodification (muscle-predominant), yeast + RNAi confirmation, LoF-by-rescue. Underpins Findings 1–5, 7–11.
35109800	<i>Novel heterozygous compound TRMT5 mutations... in a Chinese family</i>	Expands multisystem phenotype (renal/hepatic novel); documents compound-het missense variants. Findings 3, 7, 10.
20980671	<i>Mechanism of N-methylation by the tRNA m1G37 methyltransferase Trm5</i>	Defines the SAM-dependent m1G37 enzymatic reaction. Finding 2.
23887145	<i>Conservation of structure and mechanism by Trm5 enzymes</i>	m1G37 prevents ribosome frameshift errors; cross-species conservation. Findings 2, 8.
33016876	<i>Structural insights into mRNA reading frame regulation...</i>	Structural basis for m1G37 maintaining reading frame/decoding fidelity. Finding 2.
33398350	<i>A deafness-associated tRNA mutation... m1G37 modification...</i>	Demonstrates downstream cascade (↓ translation, ↓ Δψ _m , ↓ ATP, ↑ ROS, autophagy) for position-37 defects. Finding 9.
19941338	<i>Limited diagnostic value of enzyme analysis in patients with mitochondrial tRNA mutations</i>	Justifies molecular-first diagnosis; RC enzymology insensitive. Finding 5.
11282042	<i>Mitochondrial Disease</i>	Supportive/non-curative treatment framework. Finding 6.

Supporting context papers: Reviews of tRNA position-37 methylation and disease ([PMID: 40155300](#), [PMID: 38943267](#), [PMID: 37789629](#)); *TRMT5* in hepatocellular carcinoma ([PMID: 36632750](#), a distinct oncologic role of the same enzyme); and comparator mitochondrial disorders for differential diagnosis (*ACAD9* [PMID: 33204590](#); *MT-ND5* [PMID: 30587702](#); *TRMU*/reversible infantile RC deficiency [PMID: 21931168](#); *MERRF* [PMID: 21303704](#); *NARP* [PMID: 25746071](#); *MELAS* [PMID: 19198146](#)).

Limitations and Knowledge Gaps

1. **Tiny evidence base.** Fewer than ~10 patients reported. All phenotype frequencies are qualitative; no penetrance/expressivity statistics, natural-history data, survival curves, or quality-of-life measurements exist.
2. **Variant classification uncertainty.** Most *TRMT5* ClinVar entries are VUS. Without functional assays, classifying novel missense alleles is difficult, complicating diagnosis and counseling.
3. **Downstream mechanism partly inferred.** The injury cascade ($\downarrow \Delta\psi_m$, $\downarrow$ ATP, $\uparrow$ ROS, autophagy) is extrapolated from a related position-37 tRNA-modification defect ([PMID: 33398350](#)), not directly demonstrated in COXPD26 patient tissue.
4. **No whole-animal disease model.** Orthologs exist, but no published conditional knockout/knock-in recapitulating the multisystem human phenotype.
5. **Epidemiology is a computed estimate.** Carrier frequency (~1/350) and incidence (~1/490,000) derive from gnomAD pLoF counting and underestimate true incidence because hypomorphic missense alleles cause disease.
6. **No genotype–phenotype map.** It is unknown why some patients have isolated adult exercise intolerance while others have severe childhood cardiomyopathy.
7. **No therapy trials.** No treatment has been tested specifically in COXPD26; all management is extrapolated from general mitochondrial disease practice.

Proposed Follow-up Experiments / Actions

1. **Functional variant-classification pipeline.** Standardized yeast complementation and/or in vitro m1G37 methyltransferase activity assays to reclassify *TRMT5* VUS, prioritizing patient alleles (c.665T>C, c.881A>C, c.1218G>C, c.1481C>T) and producing an evidence-graded variant table.
2. **Conditional *Trmt5* animal model.** Tissue-specific (muscle, heart, CNS) mouse or zebrafish knockout to test recapitulation of muscle-predominant G37 hypomodification and combined RC deficiency, and to directly measure $\Delta\psi_m$, ATP, ROS, and autophagy.

3. **Patient iPSC and organoid models.** Differentiate patient iPSCs into cardiomyocytes, myotubes, and cortical neurons to correlate residual m1G37 modification with OXPHOS capacity and to screen candidate metabolic therapies (CoQ10, riboflavin, ROS scavengers).
4. **International registry / natural-history study.** Aggregate cases via GeneMatcher to define onset distribution, progression, survival, genotype–phenotype correlations, and standardized quality-of-life/organ outcomes.
5. **Systematic mt-tRNA modification profiling.** Modification-sensitive sequencing of patient tissues to quantify per-tRNA G37 hypomodification and link to complex-specific deficiencies.
6. **Cofactor/therapeutic pilot.** Given riboflavin responsiveness in overlapping disorders (ACAD9), evaluate cofactor supplementation in cellular models as a rational, low-risk supportive strategy.

Consensus Answer

COXPD26 (OMIM 616539; MONDO:0014684) is an ultra-rare autosomal recessive primary mitochondrial disease caused by biallelic loss-of-function variants in *TRMT5*, the nuclear-encoded enzyme that installs the m1G37 methylation on mitochondrial tRNAs; its loss causes mt-tRNA G37 hypomodification (most marked in skeletal muscle), impairing mitochondrial translation and producing a combined deficiency of respiratory-chain complexes with lactic acidosis. The phenotype is highly variable and multisystem — from childhood failure-to-thrive with hypertrophic cardiomyopathy to isolated adult exercise intolerance, plus myopathy, developmental delay, seizures, neuropathy, and GI/hepatic/renal involvement. Management is entirely supportive, as no disease-specific therapy exists.

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